



HW 4.1: In this question, we will explore the relativistic correction to the energy levels of the hydrogen atom. This is called the *fine structure correction*. Knowing that the energy of a relativistic particle is given by $E = \sqrt{p^2 c^2 + m^2 c^4}$, we can expand this expression in powers of p/mc to find the leading order relativistic correction to the energy of the electron.

- a) Write down the Hamiltonian with the leading relativistic correction to the hydrogen atom.
- b) Argue why we are justified to use non-degenerate perturbation theory to calculate the effects of this correction on the energy levels of the hydrogen atom, even though there are degeneracies in the spectrum of $\hat{H}_0$. Are the eigenstates of $\hat{H}_0$ a “good basis” for our perturbation?
- c) Using the unperturbed Schrödinger equation,

$$\hat{\mathbf{p}}^2 |n, l, m\rangle = 2m(E_n^{(0)} - V(r)) |n, l, m\rangle, \quad (1)$$

show that the leading order relativistic correction to the energy levels of the hydrogen atom can be written as

$$\Delta E_{\text{rel}} = -\frac{1}{2mc^2} \langle n, l, m | ((E_n^{(0)})^2 - 2E_n^{(0)}V(r) + V(r)^2) | n, l, m \rangle. \quad (2)$$

- d) Show that the first order correction is given by

$$\Delta E_{\text{rel}} = -\frac{(E_n^{(0)})^2}{2mc^2} \left(\frac{4n}{l + 1/2} - 3 \right), \quad (3)$$

where E_n is the non-relativistic energy of the level, n is the principal quantum number, and l is the orbital angular momentum quantum number.

- e) What order in $\alpha = \frac{e^2}{4\pi}$ is the relativistic correction? What order do you expect the next relativistic corrections to be? Why?

- f) The degeneracy of the energy levels of the hydrogen atom in ℓ was explained by the conservation of a vector called the Runge-Lenz vector:

$$\hat{\mathbf{A}} = \frac{1}{2m} (\hat{\mathbf{p}} \times \hat{\mathbf{L}} - \hat{\mathbf{L}} \times \hat{\mathbf{p}}) - k \frac{\hat{\mathbf{r}}}{r}, \quad (4)$$

where k is the positive constant in our *central* potential. With the Heisenberg picture in mind, show that the relativistic correction breaks the conservation of the Runge-Lenz vector.

(Hint: use the fact that $[\hat{H}_0, \hat{\mathbf{A}}] = 0$. You also may need $[\mathbf{V}(\mathbf{r}), \hat{\mathbf{p}}] = -i\nabla V(r)$ and $[\hat{\mathbf{r}}, \hat{\mathbf{L}}] \neq 0$).

Solutions



HW 4.1: Part of this problem is solved in Sakurai, so I refer you to the book as well. In what follows, I use natural units, so $c = 1$ and $\hbar = 1$.

a) We want to expand $E = \sqrt{p^2 + m^2}$ in powers of $p/m \ll 1$. We have $E \simeq m + \frac{p^2}{2m} - \frac{p^4}{8m^3} + \dots$. The first term is the rest mass energy which was always there to begin with. We do not worry about it because the rest energy of a particle does not alter the dynamics in non-relativistic quantum mechanics. So, we are left with the third term as the leading correction to what we already had. We can write the Hamiltonian operator as

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{rel}} = \frac{\hat{\mathbf{p}}^2}{2m} + V(r) - \frac{\hat{\mathbf{p}}^4}{8m^3}, \quad (5)$$

with $\hat{\mathbf{p}}$ the momentum operator.

b) A good basis is one in which the perturbation is diagonal. Before, we had degeneracies in ℓ and m_ℓ . Therefore, we hope that our ℓ and m_ℓ states are eigenstates of the perturbation. One easy way to see this is to ask if they are eigenstate of $\hat{\mathbf{p}}^2$, because if that is true, then they are also eigenstates of $\hat{\mathbf{p}}^4 = \hat{\mathbf{p}}^2 \hat{\mathbf{p}}^2$. We can use the unperturbed Schrödinger equation to show that this is indeed the case since

$$\hat{H}_0 |n, l, m\rangle = E_n^{(0)} |n, l, m\rangle. \quad (6)$$

Rearranging Schrödinger's equation (without the perturbation) gives

$$\hat{\mathbf{p}}^2 |n, l, m\rangle = 2m(E_n^{(0)} - V(r)) |n, l, m\rangle. \quad (7)$$

Clearly these states are eigenstates of $\hat{\mathbf{p}}^2$ and we are in luck.

c) This is a straightforward multiplication of the eigenvalues of $\hat{\mathbf{p}}^2$:

$$\hat{\mathbf{p}}^4 |n, l, m\rangle = \hat{\mathbf{p}}^2 \hat{\mathbf{p}}^2 |n, l, m\rangle = 4m^2(E_n^{(0)} - V(r))^2 |n, l, m\rangle. \quad (8)$$

and the desired result follows from the square of the parenthesis.

d) All we need here is to find the expectation values of $V(r)$ and $V(r)^2$ in the unperturbed states. These are

$$\langle n, l, m | V(r) | n, l, m \rangle = \langle n, l, m | \frac{-\alpha}{r} | n, l, m \rangle = \int_0^\infty dr r^2 R_{n,l}(r)^2 \frac{-\alpha}{r} \quad (9)$$

$$= -\alpha \left\langle \frac{1}{r} \right\rangle = -\frac{\alpha}{n^2 a_0} \quad (10)$$

where we used the fact that the spherical harmonics are normalized and the integral:

$$\left\langle \frac{1}{r} \right\rangle = \frac{1}{n^2 a_0}. \quad (11)$$

The expectation value of $V(r)^2$ is similarly given by

$$\langle n, l, m | V(r)^2 | n, l, m \rangle = \langle n, l, m | \frac{\alpha^2}{r^2} | n, l, m \rangle = \alpha^2 \left\langle \frac{1}{r^2} \right\rangle = \frac{\alpha^2}{(l + 1/2)n^3 a_0^2}. \quad (12)$$

where use the integral

$$\left\langle \frac{1}{r^2} \right\rangle = \frac{1}{n^3 a_0^2} \frac{1}{(l+1/2)} \quad (13)$$

Putting everything together, you should find

$$\Delta E_{\text{rel}} = -\frac{(E_n^{(0)})^2}{2m} \left(\frac{4n}{l+1/2} - 3 \right). \quad (14)$$

e) The relativistic correction is of order α^4 , which is an order a factor α^2 smaller than the non-relativistic energy levels of the hydrogen atom. The pattern actually repeats and the next relativistic corrections are of order α^6 , then α^8 , and so on. One way to see this is to note that α is essentially the velocity of the electron in the hydrogen atom, so we are doing an expansion in powers of v/c (in natural units, $c = 1$ so v is dimensionless and smaller than one always). In other words, the semi-relativistic expansion of $\sqrt{p^2 + m^2}$ is an expansion in powers of v :

$$E = m\sqrt{1 + \frac{p^2}{m^2}} = m\sqrt{1 + v^2} \simeq m + \frac{1}{2}mv^2 - \frac{1}{8}mv^4 + \frac{1}{16}mv^6 - \dots \quad (15)$$

Another way to see it is to look at the expectation value of

$$\langle n, l, m | \hat{\mathbf{p}}^6 | n, l, m \rangle = 8m^4 \langle n, l, m | (E_n^{(0)} - V(r))^3 | n, l, m \rangle, \quad (16)$$

which will contain terms of the form $\langle n, l, m | V(r)^3 | n, l, m \rangle$, which is of order α^6 . One can actually calculate it with the following relation:

$$\langle n, l, m | V(r)^3 | n, l, m \rangle = -\alpha^3 \left\langle \frac{1}{r^3} \right\rangle = -\frac{\alpha^3}{l(l+1/2)(l+1)n^3 a_0^3} = -\frac{\alpha^6 m_e^3}{l(l+1/2)(l+1)n^3}. \quad (17)$$

Beyond this order, the integrals become nastier, but the procedure is the same.

f) In the Heisenberg picture the time-evolution of an operator with no explicit time dependence is

$$\frac{d\hat{\mathbf{A}}}{dt} = i[\hat{H}_0 + \hat{H}_{\text{rel}}, \hat{\mathbf{A}}]. \quad (18)$$

We are given (and know) that

$$[\hat{H}_0, \hat{\mathbf{A}}] = 0. \quad (19)$$

Therefore the relativistic correction changes the conservation law according to

$$\frac{d\hat{\mathbf{A}}}{dt} = i[\hat{H}_{\text{rel}}, \hat{\mathbf{A}}] = -\frac{i}{8m^3} [\hat{\mathbf{p}}^4, \hat{\mathbf{A}}]. \quad (20)$$

So it suffices to show $[\hat{\mathbf{p}}^4, \hat{\mathbf{A}}] \neq 0$ to show that $\hat{\mathbf{A}}$ is not conserved when the relativistic correction is included. Note also that since $\mathbf{p}^2$ commutes with itself, we can write

$$[\hat{\mathbf{p}}^4, \hat{\mathbf{A}}] = \hat{\mathbf{p}}^2 [\hat{\mathbf{p}}^2, \hat{\mathbf{A}}] + [\hat{\mathbf{p}}^2, \hat{\mathbf{A}}] \hat{\mathbf{p}}^2, \quad (21)$$

So, one plan of attack is to target $[\hat{\mathbf{p}}^2, \hat{\mathbf{A}}]$ and that it is non-zero (technically, we must show it does not anticommute with $\hat{\mathbf{p}}^2$, but that should be straightforward).

First, note that

$$\left[\hat{H}_0, \hat{\mathbf{A}} \right] = \left[\frac{\hat{\mathbf{p}}^2}{2m}, \hat{\mathbf{A}} \right] + [V(r), \hat{\mathbf{A}}] = 0 \quad \Rightarrow \quad [\hat{\mathbf{p}}^2, \hat{\mathbf{A}}] = -2m [V(r), \hat{\mathbf{A}}]. \quad (22)$$

This already shows that $[\hat{\mathbf{p}}^2, \hat{\mathbf{A}}]$ is not separately zero (the kinetic and potential terms cancel only in the combination $\hat{H}_0$).

Since $V(r)$ is a function of r only, it commutes with $\hat{\mathbf{r}}/r$ and with $\hat{\mathbf{L}}$ (central potential), i.e. $[V(r), \hat{\mathbf{r}}/r] = 0$ and $[V(r), \hat{\mathbf{L}}] = 0$.

i

If this is not obvious to you, recall that the conservation of angular momentum in the Heisenberg picture is given by $d\hat{\mathbf{L}}/dt = i[\hat{H}_0, \hat{\mathbf{L}}] = 0$. Since the kinetic term of $\hat{H}_0$ commutes with $\hat{\mathbf{L}}$ (that is, $[\hat{\mathbf{p}}^2/2m, \hat{\mathbf{L}}] = 0$), the potential term must also commute with $\hat{\mathbf{L}}$ to ensure the conservation of angular momentum.

Also, it should also be clear that $[\hat{\mathbf{p}}^2, \hat{\mathbf{L}}] = 0$ because angular momentum is a generator of rotations and $\hat{\mathbf{p}}^2$ is a scalar under rotations. One can also see this by explicit calculation using $[r_i, p_j] = i\delta_{ij}$ and the fact that an antisymmetric tensor contracted with a symmetric tensor is zero.

Thus only the momentum in the first term contributes:

$$[V, \hat{\mathbf{A}}] = \frac{1}{2m} \left([V, \hat{\mathbf{p}}] \times \hat{\mathbf{L}} - \hat{\mathbf{L}} \times [V, \hat{\mathbf{p}}] \right) = -\frac{i}{2m} \left((\nabla V) \times \hat{\mathbf{L}} - \hat{\mathbf{L}} \times (\nabla V) \right). \quad (23)$$

For a radial potential $V(r)$, $\nabla V = V'(r) \hat{\mathbf{r}}$. Crucially, $\hat{\mathbf{L}}$ does *not* commute with $\hat{\mathbf{r}}$ (as the hint emphasizes):

$$[\hat{L}_k, \hat{r}_l] = [\epsilon_{ijk} \hat{r}_i \hat{p}_j, \hat{r}_l] = \epsilon_{ijk} ([\hat{r}_i, \hat{r}_l] \hat{p}_j + \hat{r}_i [\hat{p}_j, \hat{r}_l]) = i \epsilon_{ijk} \delta_{jl} \hat{r}_i = i \epsilon_{ilk} \hat{r}_i \neq 0, \quad (24)$$

so the operator $(\nabla V) \times \hat{\mathbf{L}} - \hat{\mathbf{L}} \times (\nabla V)$ is not zero and therefore

$$[V(r), \hat{\mathbf{A}}] = [\hat{\mathbf{p}}^2, \hat{\mathbf{A}}] \neq 0 \quad (25)$$

by Eq. (22). The commutator of $\hat{\mathbf{p}}^2$ with $(\nabla V) \times \hat{\mathbf{L}} - \hat{\mathbf{L}} \times (\nabla V)$ is actually quite nasty, but we do not worry about a possible anti-commutation, and we can conclude that $[\hat{\mathbf{p}}^4, \hat{\mathbf{A}}] \neq 0$.